Problem 2: Finding Complexity using Counter method

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
  if(n==1)
  {
    printf("*");
  }
  else
  {
   for(int i=1; i<=n; i++)
   {
     for(int j=1; j<=n; j++)
     {
       printf("*");
```

```
        printf("*");
        break;
      }
    }
  }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:(penalty regime: 0 %)

```
#include<stdio.h>
void func(int n)
{
   int counter = 0;
   if(n==1)

   {
      counter++;
   }
   else
   {
    counter++;}
    for(int i=1; i<=n; i++){
       counter++;
       counter++;
```

```
      for(int j=1; j<=n; j++){
         counter++;
         counter++;
         counter++;
         break;}}
         counter++;
  printf("%d",counter);
}



int main(){
   int n;
   scanf("%d",&n);
   func(n);
   return 0;
}
```

	Input	Expected	Got	
	2	12	12	
	1000	5002	5002	
	143	717	717	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.